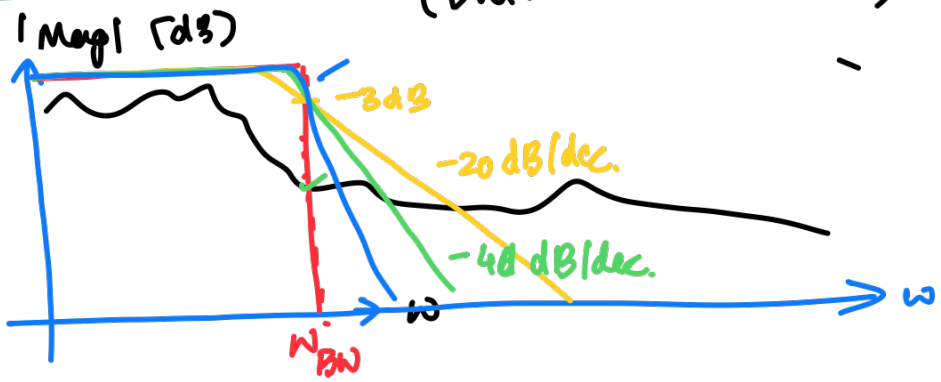
Lecture #32

(Butterworth filter)



$H(s)$

$20\log_{10}|H(j\omega)| = 10\log_{10}(|H(j\omega)|^2)$

$$H(s) = \frac{N(s)}{D(s)} = \frac{1}{D(s)}$$

$$= \frac{1}{\alpha_0 s^0 + \alpha_1 s^1 + \alpha_2 s^2 + \ldots + \alpha_n s^n + \ldots}$$

$$= \frac{1}{\sum \alpha_{2k} s^{2k} + \sum \alpha_{2p+1} s^{2p+1}}$$

$$H(j\omega) = \frac{1}{\sum \alpha_{2k} (j\omega)^{2k} + \sum \alpha_{2p+1} (j\omega)^{2p+1}}$$

$$|H(j\omega)|^2 = \frac{}{\sum \alpha'_{2k} \omega^{2k} + \sum \alpha'_{2p+1} (\omega^2)^p}$$

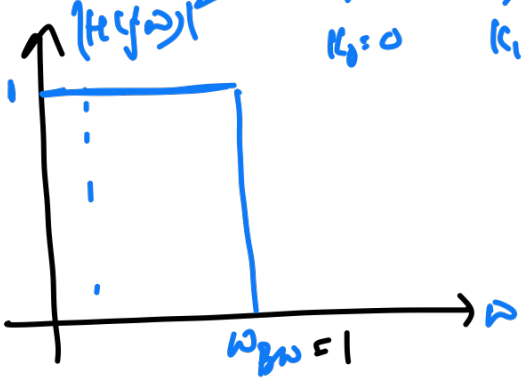
$$|\alpha_0 + j\alpha_1\omega|^2 = \alpha_0^2 + \alpha_1^2\omega^2$$

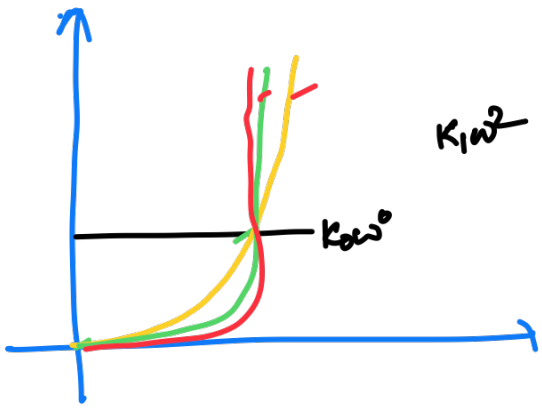
$$\alpha_0 + j(\alpha_1\omega - \alpha_2\omega^3) = \alpha_0^2 + \alpha_1^2\omega^2 + \alpha_2^2\omega^6 - 2\alpha_1\alpha_2\omega^4$$

$$|H(j\omega)|^2 = \frac{1}{1 + K(\omega^2)} \quad \checkmark$$

$$|D(j\omega)|^2 = 1 + K_0\omega^0 + K_1\omega^2 + K_2\omega^4 + \cdots + K_m\omega^{2m} + \cdots + K_n\omega^{2n} + \cdots$$

$K_0 = 0$ $\quad K_1 = 0$ $\quad K_2 = 0$ $\quad K_m \neq 0$





$$|H(j\omega)|^2 = \frac{1}{1+\omega^{2m}}$$

$$\frac{\partial^K |H(j\omega)|^2}{\partial(\omega^2)^K} = 0$$

"maximally flat"

$$|H(j\omega)|^2 = \frac{1}{1+\left(\frac{\omega}{\omega_0}\right)^{2m}} \checkmark \quad ; \quad \omega_{BW} = \omega_0$$

$$= H(j\omega) \times \left(H(j\omega)\right)^*$$

$$= H(j\omega) \times H(-j\omega)^*$$

$$= H(s) \times H(-s).$$

$$s = j\omega \Rightarrow |H(j\omega)|^2 = H(s) \times H(-s) = \frac{1}{1+\left(\frac{s}{j} \times \frac{1}{\omega_0}\right)^{2m}}$$

$$= \frac{1}{1+(-j)^{2m}\left(\frac{s}{\omega_0}\right)^{2m}}$$

$$1 + (-j)^{2m}\left(\frac{s}{\omega_0}\right)^{2m} = 0$$

$$\left(\frac{s}{\omega_0}\right)^{2m} = \frac{-1}{(-j)^{2m}} = \frac{e^{j(2q+1)\pi}}{(-j)^{2m}}$$

if m is even

$$\left(\frac{s}{\omega_0}\right)^{2m} = \frac{e^{j(2q+1)\pi}}{1}$$

$(-j)^{2m}$ $\xrightarrow{m=0}$ 1, $\xrightarrow{m=2}$ 1

$((-j)^2)^m = (-1)^m$

if m is odd

$$\left(\frac{s}{\omega_0}\right)^{2m} = \frac{e^{j(2q+1)\pi}}{-1}$$

Ex: $m = 2$

$$\frac{s}{\omega_0} = e^{j(2q+1)\pi/4} \quad ; \quad q = 0, 1, 2, 3$$

$$= e^{j\pi/4}, \; e^{j3\pi/4}, \; e^{j5\pi/4}, \; e^{j7\pi/4}$$

R.H.P L.H.P ✓ L.H.P ✓ R.H.P

$H(s)$: $s = \omega_0\left(\frac{-1}{\sqrt{2}} + \frac{j}{\sqrt{2}}\right), \; \omega_0\left(\frac{-1}{\sqrt{2}} - \frac{j}{\sqrt{2}}\right)$

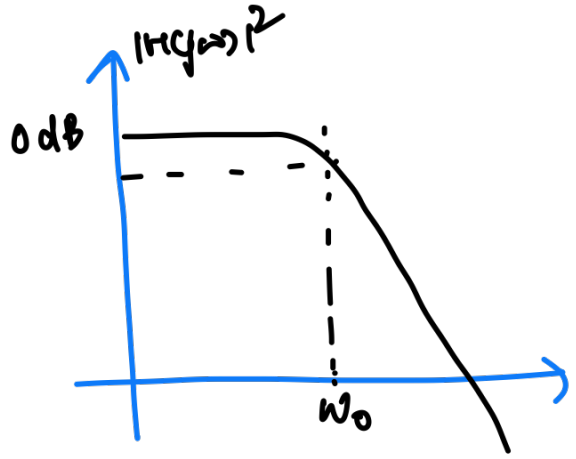
$$H(s) = \frac{1}{\left(s + \frac{\omega_0}{\sqrt{2}} - j\frac{\omega_0}{\sqrt{2}}\right)\left(s + \frac{\omega_0}{\sqrt{2}} + j\frac{\omega_0}{\sqrt{2}}\right)}$$

$$= \frac{}{\left(s + \frac{\omega_0}{\sqrt{2}}\right)^2 + \frac{\omega_0^2}{2}}$$

$$= \frac{}{s^2 + \frac{\omega_0^2}{2} + \frac{2s\omega_0}{\sqrt{2}} + \frac{\omega_0^2}{2}}$$

$$= \frac{}{s^2 + \sqrt{2}\omega_0 s + \omega_0^2}$$

$$= \frac{1}{\frac{s^2}{\omega_0^2} + \sqrt{2}\frac{s}{\omega_0} + 1}$$



$$\frac{s^2}{\omega_0^2} + \frac{s}{\omega_0 Q_p} + 1$$

$$Q_p = \frac{1}{\sqrt{2}}$$

- $Q_p < \frac{1}{2}$, real poles.
- $\frac{1}{2} < Q_p < \frac{1}{\sqrt{2}}$ complex poles, no peaking
- $Q_p > \frac{1}{\sqrt{2}}$ peaking.

$$\left(\frac{s}{\omega_0}\right)^{2m} = \frac{e^{j(2q+1)\pi}}{-1} = e^{j2q\pi}$$

Ex: $m = 3$

$$\left(\frac{s}{\omega_0}\right) = -1\, e^{j(2q+1)\pi/6} \quad ; \quad q = 0, 1, 2, 3, 4, 5$$

$$\frac{s}{\omega_0} = -1\left(e^{j\pi/6}\right), -1\left(e^{j\pi/2}\right), -1\left(e^{j5\pi/6}\right),$$

$$-1\left(e^{j7\pi/6}\right), -1\left(e^{j9\pi/6}\right), -1$$

$$\frac{s}{\omega_0} = e^{j\frac{2q\pi}{6}\times 3}$$

$$= e^{j0}, e^{j\pi/3}, e^{j2\pi/3}, e^{j\pi}, e^{j4\pi/3}, e^{j5\pi/3}$$

L.H.P R.H.P L.H.P R.H.P L.H.P L.H.P

$$\left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) \quad -1 \quad \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right)$$

$$s = -\omega_0, \quad +\omega_0\left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right), \quad \omega_0\left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right)$$

$$H(s) = \frac{}{(s+\omega_0)\left(\left(s+\frac{\omega_0}{2}\right)^2 + \frac{3\omega_0^2}{4}\right)}$$

$$= \frac{}{(s+\omega_0)\left(s^2 + \frac{\omega_0^2}{4} + \omega_0 s + \frac{3\omega_0^2}{4}\right)}$$

$$= \frac{}{(s+\omega_0)\left(s^2 + \omega_0 s + \omega_0^2\right)}$$

$$= \frac{1}{\left(\frac{s}{\omega_0}+1\right)\left(\frac{s^2}{\omega_0^2} + \frac{s}{\omega_0} + 1\right)} \checkmark$$

$\frac{s^2}{\omega_0^2} + \frac{s}{\omega_0 Q} + 1$

